

CURRICULUM VITAE

Yiwei Ou, Ph.D.

School of Architecture and Planning, The University of Auckland, New Zealand

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EDUCATION

Ph.D. in Architecture (with University of Auckland Doctoral Scholarship), Dec. 2021 – Sep. 2026

The University of Auckland, Auckland, New Zealand

Dissertation: “New Central Spaces of Consumption: Understanding Urban Inversion in China’s Ultra-modern Precinct Malls with the Methods of Space Syntax and Machine Learning”

Advisors: Prof. [Manfredo Manfredini](#), Prof. [Kaiqi Zhao](#), Dr. [Ferdinand Oswald](#)

GPA: 8.875/9.0

Artificial Intelligence Graduate Certificate, Mar. 2023 – Dec. 2024

Stanford University, Stanford, CA, USA

Courses: [Machine Learning \(CS229\)](#), [Natural Language Processing with Deep Learning \(CS224N\)](#), [Deep Learning for Computer Vision \(CS231N\)](#), [Machine Learning with Graphs \(CS224W\)](#)

GPA: 4.075/4.3

Master of Architecture (with First Class Honors), Feb. 2020 – Oct. 2021

The University of Auckland, Auckland, New Zealand

Dissertation: “Public Space and Consumption in the Mobile Internet Era”

Advisor: Prof. [Manfredo Manfredini](#)

GPA: 9.0/9.0

Master of Architecture (with Merit Graduate Honors), Sep. 2018 – Jun. 2021

Hunan University, Changsha, China

Dissertation: “Research on the Spaces of Urban Commercial Centers Based on the Theory of ‘Inverted City’”

Advisors: Prof. [Canhong Qiu](#), Prof. [Yicheng Yin](#)

GPA: 3.83/4.0

MIT Summer Program, Aug. 2019

Massachusetts Institute of Technology, Cambridge, MA, USA

Project: “Game Design for Interacting with Urban Public Space”

Best Project Award

Bachelor of Architecture (with Merit Graduate Honors), Sep. 2013 – Jun. 2018 (5-year degree)

Hunan University, Changsha, China

Thesis: “Historical Block Protection and Renewal & Industrial Building Reconstruction and Reuse Design”

GPA: 3.44/4.0

RESEARCH EXPERIENCE

Research Assistant, 02/2024~09/2026

The University of Auckland, New Zealand (PIs: [Farzaneh Haghighi](#), [Manfredo Manfredini](#))

- **Project:** Mapping New Geographies: Investigating the 2022 Parliament Grounds Protests in Wellington, NZ
- **Description:** Applied computer vision and machine learning methods to large-scale social media datasets, including training models to detect and classify urban landmarks, mapping the geospatial distribution of urban facilities, and conducting spatial-temporal analysis of digitally mediated urban activity.

Research Assistant, 10/2023~06/2024

The University of Auckland, New Zealand (PI: [Elham Bahmanteymouri](#))

- **Project:** [Cities and Digital Platforms](#)
- **Description:** Conducted literature review and data collection examining the impacts of short-term-rental digital platforms on land use, urban economics, and urban development patterns across different geographic contexts.

Research Assistant, 11/2023~05/2024

The University of Auckland, New Zealand (PI: [Paola Boarin](#))

- **Project:** [Future Cities Research Centre](#)
- **Description:** Maintained and updated the Future Cities Research Hub website, supporting the dissemination and communication of research activities and outputs.

TEACHING AND SUPERVISING EXPERIENCE

Co-supervisor (with Dr. [Xiaobin Ren](#), Prof. [Kaiqi Zhao](#)), 06/2024~present

- **Thesis:** Master of Science (MSc) in Computer Science
- **Description:** Co-supervising a master's thesis on the development of a multimodal street-level visual place recognition (VPR) dataset and benchmark for urban environments in Chinese commercial urban districts, contributing to data collection protocols, research design, and thesis guidance.

Co-supervisor (with Prof. [Manfredo Manfredini](#)), 02/2023~02/2024

- **Thesis:** Master of Architecture (Professional)
- **Description:** Co-supervised a master's thesis, providing guidance on thesis framework development, research question formulation, architectural theory, research methodology, and academic writing.

Invited Course Facilitator, 08/2025

Stanford University (Online), USA

- **Course:** [XCS231N: Deep Learning for Computer Vision](#)
- **Description:** Invited to serve as a course facilitator for Stanford's professional-level computer vision course, with responsibilities including supporting course delivery, student engagement, and learning facilitation.

Teacher, Co-instructor, 02/2024~10/2024

The University of Auckland, New Zealand

- **Course:** [Academic Career Advancement Programme](#) (ACAP) 2024
- **Description:** Delivered lectures and facilitated seminars addressing key aspects of academic life and fostering interdisciplinary connections among academics across the University of Auckland.

Graduate Teaching Assistant, 09/2018~12/2019

Hunan University, China

- **Courses:** Architectural Design, Theory of Urban Design, Introduction to Sciences of Human Settlement, Chinese and Foreign Urban Construction History, Historic City Protection
- **Description:** Supported course development and delivery, including assignment and examination preparation, student project mentoring, and course administration for graduate-level courses.

AWARDS AND HONORS

- Best PhD Thesis Award on Public Space Research—Shortlisted (Top 9), City Space Architecture, July 2026
- Best Master Thesis Award on Public Space Research—Honourable Mention, City Space Architecture, June 2024
- University of Auckland Academic Career Advancement Award, December 2023
- University of Auckland Distinguished Graduate Award, October 2023
- University of Auckland Doctoral Scholarship, December 2021
- First Class Honors of Master of Architecture, The University of Auckland, Oct 2021
- Outstanding Graduate Award for Innovation and Entrepreneurship of Hunan Province (Postgraduate), Jun 2021
- Outstanding Graduate Award for Innovation and Entrepreneurship of Hunan University (Postgraduate), Jun 2021
- Graduate Merit Scholarship, Hunan University, Dec 2019
- First-class Academic Scholarship, Hunan University, Oct 2018; Oct 2019
- Outstanding Graduate Award for Innovation and Entrepreneurship of Hunan Province (Undergraduate), Jun 2018
- Outstanding Graduate Award for Innovation and Entrepreneurship of Hunan University (Undergraduate), Jun 2018
- First-class Academic Scholarship, Hunan University, Oct 2017
- Undergraduate Merit Scholarship, Hunan University, Dec 2016
- Excellence Award of the 6th National Green Buildings Design Competition, China, Nov 2016
- National Scholarship, China, 2014-2016

PUBLICATIONS (*corresponding author)

1. **Ou, Y. ***, Cheung, C. C., Ang, J. Y., Ren, X., Sun, R., Gao, G., Zhao, K., & Manfredini, M. (2026). Urban-ImageNet: A Large-Scale Multi-Modal Dataset and Evaluation Framework for Urban Space Perception. arXiv:2605.09936. <https://arxiv.org/abs/2605.09936>. [[Dataset](#), [Code](#)]
2. **Ou, Y. ***, & Manfredini, M. (2026) Futurible Heterotopias of Phygital Spectacle: Understanding How Digital Technologies Intensify Spatial Segregation in Quasi-Public Commercial Centres, The Journal of Public Space, 10(2), pp. 47-76. <https://doi.org/10.32891/jps.v10i2.1882>

3. Guo, Z., Manfredini, M., **Ou, Y.**, & Jiang, J. (2026). Rationalise, Distract, and Detour: Decoding the Spatial Logic of the Hetero-inverted Consumption Space in Urban China. 15th International Space Syntax Symposium, Johor Bahru, Malaysia, 8–12 June 2026.
4. **Ou, Y. ***, Ren, X., Sun, R., Gao, G., Jiang, Z., Zhao, K., & Manfredini, M. (2025). MMS-VPR: Multimodal Street-Level Visual Place Recognition Dataset and Benchmark. <http://arxiv.org/abs/2505.12254ss> [Dataset, Code]
5. **Ou, Y. ***, & Manfredini, M. (2024). Discipline and ‘Consumer’: Spatial Configuration Ringiness in Open-Plan Commercial Centres in China. The 14th International Space Syntax Symposium, Nicosia, Cyprus, 23, June 2024.
6. **Ou, Y. ***, & Manfredini, M. (2024). New Digital Technologies in Quasi-Public Commercial Centres: Extended Hyper-Reality, Illusion and Augmented Social Interactions. International Conference on Past Present and Future of Public Space 2024 (PPFPS), Bologna, Italy, 75, June 2024.
7. **Ou, Y. ***, Zhou, X., Tu, H., Chen, Y., & Jia, X. (2023). Evaluating spatial justice through the analysis of the relationship between house prices and the accessibility to urban infrastructures with a big data approach. The 11th International Conference of the Arab Society for Computation in Architecture, Art and Design (ASCAAD), Amman, Jordan, 441, November 2023.
8. Chen, Y., Jia, X., Tu, H., **Ou, Y.**, & Zhou, X. (2023). City diversity: how do architectural uses, ages, and styles affect the public? – A case study in Manhattan through social media data. The 11th International Conference of the Arab Society for Computation in Architecture, Art and Design (ASCAAD), Amman, Jordan, 463, November 2023.
9. **Ou, Y. ***. (2023). A comparative study on the spatial configurations and urban connectivity of the paradigms of enclosed and precinct malls: The cases of Changsha IFS and Chengdu Taikoo Li in China. The 2nd International Space Syntax PhD Conference (SS2PhD), June 2023.
10. **Ou, Y. ***, Manfredini, M., Gao, G., Sun, R., & Su, Q. (2023). Sociospatial Wellbeing in Urban Commercial Centres: An Analysis of Abstraction and Relationality of Novel Mall Spatialities in the Mobile Internet Era. The 6th International Conference on Indonesian Architecture and Planning (ICIAP), Lecture Notes in Civil Engineering, 2023, vol 334, p. 557-578. Springer, Singapore. https://doi.org/10.1007/978-981-99-1403-6_37
11. Su, Q., Manfredini, M., Sun, R., & **Ou, Y.** (2023). Opportunities and Challenges for Rural Migrant Workers in Villages in the City: A Perspective of Translocal Reproduction of Labour and Re-Commoning. The 6th International Conference on Indonesian Architecture and Planning (ICIAP), Lecture Notes in Civil Engineering, 2023, vol 334, p. 151-162. Springer, Singapore. https://doi.org/10.1007/978-981-99-1403-6_12
12. **Ou, Y. ***, & Manfredini, M. (2022). The Configurational Analysis of the Ultra-Modern Shopping Centre in the Urban Restructuring of the Mediatized Era: A Study of Ultra-Modern Centre Based on the Method of Space Syntax. The 13th International Space Syntax Symposium (SSS13), Bergen, Norway, 1, June 2022.
13. Gao, G., Chen, F., & **Ou, Y.** (2022). Study on the Spatial Design of Large Independent Bookstores: Taking Changsha Meixi Master as an Example, China Science Paper Online, 202203-248, March 2022.
14. **Ou, Y. ***, & Manfredini, M. (2020). Public Space and Consumption in the Mobile Internet Era - A Study on the Heterotopic Inversion of Ultra-Modern Shopping Malls. The 13th International Forum on Urbanism (IFoU), p. 390-409. October 2020.

MANUSCRIPTS IN PREPARATION

1. **Ou, Y.**, Manfredini, M., Gao, G. Multimodal Urban Perception Analysis.
2. **Ou, Y.** Uncovering Spatial Injustice through Machine Learning: A Feature Importance Analysis of Supermarket Accessibility Determinants in Dutch Neighborhoods.
3. **Ou, Y.**, Manfredini, M. Passive E-participation in Urban Design: A Novel Machine Learning Approach to Extract Public Attitudes Towards Different Types of Urban Spaces with Social Media Big Data.
4. **Ou, Y.**, Cheung, C., Ang, J. GeoAI-Powered Analysis of Urban Space Perceptions: Harnessing Advanced Computer Vision Models and Social Media Big Data for Smart City Planning.
5. **Ou, Y.**, Ang, J., Cheung, C. Detecting Urban Landmarks through Unsupervised Learning and Big Data of Social Media Images: A Case Study of Chengdu Central Business Area.
6. **Ou, Y.**, Bischoff, J. Predicting Metro-Station-Community House Prices with Graph Neural Networks: A Case Study of Shanghai and Beijing Subway Networks.
7. **Ou, Y.**, Wu, J. A Big Data Tool to Evaluate the Urban Spatial Justice of the Accessibility to Shopping Facilities: Evidence from National-wide Neighborhoods in the Netherlands.

INVITED TALK

1. “AI/CV for Spectacular Simulation” at Shanghai Academy of Fine Arts, Shanghai University, October 2025.
2. “GeoAI and Social Media: A Novel Framework for Analyzing Public Space Perceptions” (for visiting faculty and students from School of Architecture, Zhengzhou University), at The University of Auckland, August 2025.
3. “Detect Public Perceptions of Malls with Social Media Images and Machine Learning Methods” (for

visiting faculty from School of Architecture, Wuhan University), at The University of Auckland, May 2025.

4. “Beyond Distance: Machine Learning Approaches to Understanding Urban Spatial Justice” at School of Architecture and Planning, The University of Auckland, New Zealand, February 2024.
5. “The Configurational Analysis of the Ultra-Modern Shopping Centre in the Urban Restructuring of the Mediatized Era” at Shanghai Academy of Fine Arts, Shanghai University, China, June 2022.

SERVICE TO PROFESSION

Journal Article Reviewer

- [Sustainable Cities and Society](#) (Cite Score=22.3, IF=13.3), 02/2024~Present
- [Computers, Environment and Urban Systems](#) (Cite Score=17.3, IF=8.9), 01/2024~Present

Conference Proceeding Reviewer

- [2025 Interdisciplinary Symposium on the Sustainable Development Goals](#), 07/2025~09/2025
- [International Union of Architects \(UIA\) World Congress](#), 10/2022
- [International Space Syntax Symposium](#), 04/2022~Present

Expert Research Evaluation

- Invited Expert Contributor for [Research Novelty Evaluation, UK Metascience Unit & University of Sussex](#), 2026

Committee Member

- Early-Career Advisor Board of [City Space Architecture](#), 02/2022~Present
- [Spatiotemporal Innovation Lab for Big Data](#) at Harvard University, 12/2022~Present
- New Zealand Chinese Scientists Association (NZCSA), 08/2023~Present
- [New Zealand Association of Scientists](#) (NZAS), 08/2023~Present
- [New Zealand Institute of Architects](#) (NZIA), 01/2023~Present
- [Architectural Society of China](#) (ASC), 05/2018~Present

CERTIFICATIONS

Machine Learning & Data Science

- [Artificial Intelligence Graduate Certificate](#), Stanford University, 12/2024
- [Neural Networks and Deep Learning](#), DeepLearning.AI, 01/2023
- [Machine Learning](#), Stanford Online, 07/2022
- [Code in place](#), Stanford University, 06/2023
- [Harvard Spatiotemporal Innovation Workshop: Social Media Sentiment Analysis](#), Future Data Lab & Center for Geographic Analysis, Harvard University, 12/2022
- [Machine Learning 101 on NeSI](#), New Zealand eScience Infrastructure (NeSI), 03/2024
- [Machine Learning Workshop](#), The University of Auckland, 04/2023
- [Data Carpentry Workshop](#), The University of Auckland, 08/2022
- [Neural Networks for Predicting Subjective Design Evaluation](#), DigitalFUTURES, 07/2022
- [International Workcamp for Healthy Cities Science and Planning 2023: Analysis techniques for healthy city data based on R language](#), Tongji University & Hunan University, 05/2023

Academic Workshops

- [Academic Career Advancement Programme](#), The University of Auckland, 10/2024
- [UoA Critical Edge Workshop](#), The University of Auckland, 09/2024
- [PhD Peer Learning Workshops](#), The University of Auckland, 08/2023
- [Southeast University Summer School 2022: Frontiers of International Planning Theories and Methods](#), School of Architecture, Southeast University, 09/2022

OTHER ACHIEVEMENTS

Senior Group Facilitator, 06/2026

The 20th New Zealand Annual [Aspiring Leaders' Forum \(ALF\) on Faith and Values](#), Wellington, New Zealand

National Nominated Delegate, 08/2023

The 17th New Zealand Annual [Aspiring Leaders' Forum \(ALF\) on Faith and Values](#), Wellington, New Zealand

Academic Event Organizer, 01/2023~01/2024

[New Zealand Chinese Postgraduate Society](#), Auckland, New Zealand

Vice President, 09/2018~10/2019

Graduate Students Council of the School of Architecture and Planning, Hunan University, Changsha, China